



# OptiTriage

Problem Statement: Edge Native Optical  
Triage Engine.

Domain: Healthcare

Team Assembly Coders

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HACK  
MATRIX

# The Problem

## "Blind" Telemedicine

Virtual consults lack objective vitals, forcing doctors to prescribe based on visual guesswork

## ER Triage Bottlenecks

Overwhelmed waiting rooms delay manual vitals, letting critical patients deteriorate undetected.

## Cloud & Bias Risks

Cloud video streaming creates HIPAA risks, while basic tools fail on darker skin tones.

Who is affected?

Telemedicine Patients & ER Visitors → Physicians & Triage Nurses → Healthcare Systems & Rural Clinics

Problem Statement ID:

PS-XX



# WHY THIS PROBLEM EXISTS

## Hardware Gap

Telemedicine lacks physical diagnostic sensors.

## Manual Triage

Overwhelmed ERs cause dangerous wait times

## Flawed Tech

Cloud processing risks privacy, and basic algorithms exhibit skin-tone bias.

V I

CORE PROBLEM – USER CONSEQUENCE – REAL  
Diagnostic Blind Spot > Doctors Rely on Visual Guesswork > IMPACT  
Silent Patient Deterioration

Current Approach: Standard Video Chats & Manual Triage →  
Requires Physical Hardware/Apps → Severe Wait Times &  
HIPAA Risks → Fatal Bottlenecks





# OUR SOLUTION

## INTRODUCING OptiTriage

**OptiTriage** enables **physicians and triage nurses** to **access real-time, objective physiological data** by **converting any standard webcam into a clinical-grade vital sign monitor using edge-native compute.**

Patient clicking a secure SMS link  
↓  
30-second standard webcam face scan  
↓  
Browser-based HTML5 Canvas & MediaPipe  
↓  
WebGPU signal amplification & CHROM algorithm  
↓  
BPM, HRV, Respiratory Rate & NEWS2 Score  
↓  
Immediate, contactless triage with zero hardware

### Core capabilities

#### Capability Value

- 01 Edge-Native Compute (Zero-Cloud): Video never leaves local memory, ensuring zero bandwidth costs, infinite scalability, and inherent HIPAA/GDPR compliance.
- 02 Health Equity via CHROM Algorithm: Dynamically cancels out melanin absorption bias and sweat reflections to guarantee clinical accuracy across all skin tones.
- 03 Frictionless Mass: Adoption Requires no proprietary sensors, no app installations, and no waiting—patients just click a link and scan.





# PROTOTYPE / PRODUCT EXPERIENCE

## Prototype Screenshots

1

## Prototype Screenshots

2

Right: User journey

01 — INPUT

Patient clicks a secure SMS link for a 30-second webcam scan.



02 — PROCESS

Edge-native compute processes the video entirely locally using the CHROM algorithm.



03 — OUTPUT

System extracts accurate BPM, HRV, Respiratory Rate.



04 — ACTION

A lightweight data payload instantly alerts the doctor's dashboard for immediate triage.



# WHAT MAKES US DIFFERENT?

## WHY THIS

## APPROACH?

Traditional computer vision and telehealth rely on heavy cloud processing and biased algorithms. We engineered a strictly edge-native, equitable alternative.

### Existing Approach

### OptiTriage

Requires App/Hardware



Manual ER Triage Bottlenecks



Algorithmic Skin-Tone Bias



Cloud Video Privacy Risks



Scalability Limited





# TECHNICAL ARCHITECTURE

Tech stack:

Frontend:

React / Vite

Backend:

Node.js / ExpressJS

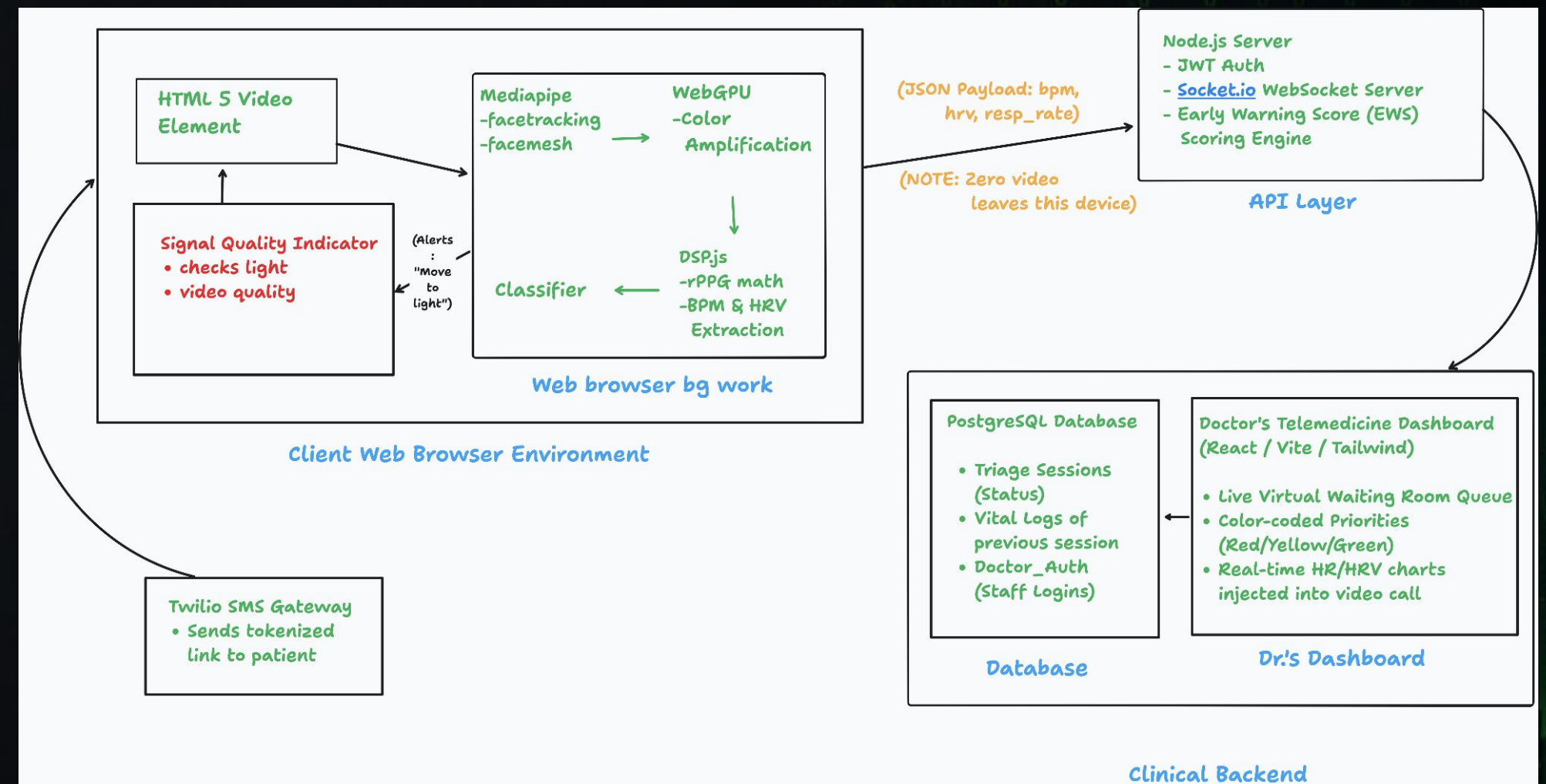
Database:

Supabase

AI/ML:

MediaPipe (Face Tracking) / WebGPU  
(Color Amplification) / CHROM Algorithm

Example Workflow  
Image



# FEASIBILITY + IMPACT

## FEASIBILITY

Technical-

Standard Webcams & Browsers

MediaPipe Face Mesh & WebGPU

100% Local Edge Compute

Operational-

Zero proprietary hardware cost

Frictionless SMS link deployment

Hackathon MVP-

Live webcam vitals pushed to

Doctor's Dashboard

## IMPACT

Who benefits?

Telemedicine Docs & ER Patients

What improves?

Instant, contactless triage vitals

Why does it matter?

Prevents silent patient deterioration

Then a bottom progression:

Prototype → Hackathon MVP → ER

Waiting Room Pilot → Global Rural

Scale





# IMPLEMENTATION ROADMAP

## WHAT HAPPENS NEXT?

### NOW

- Edge-Native Pipeline Built
- CHROM Algorithm Integrated
- Live Doctor's Dashboard Active

### NEXT

#### MVP

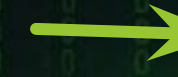
- Clinical Validation (NEWS2)
- Skin-Tone Equity Testing
- Signal Quality UI Refinement

### LATER

- ER Waiting Room Kiosk Pilots
- Telehealth Platform Integration
- Provider Analytics Dashboard

### SCALE

- Global Rural Telemedicine Rollout
- NICU & Burn Victim Monitoring
- Millions of Users at Near-Zero Cost



# TEAM + EXECUTION

| Shrikar Shelar    | Mayur Kaithwas        | Sadhak Shinde     | Harsh Shedge        |
|-------------------|-----------------------|-------------------|---------------------|
| Edge Compute Lead | Vision Engineer       | Backend Architect | Clinical UI/UX Lead |
| WebGPU & Wasm     | MediaPipe & rPPG Math | WebSockets & DB   | React & Dashboards  |

Our combined expertise spanning edge-native computing, real-time signal processing, and secure full-stack architecture perfectly positions us to build and scale OptiTriage without relying on traditional cloud bottlenecks.

